

A Flux-Free Finite-State Trigger for Onboard Solar Energetic Particle Onset Detection from Raw Counts, Validated Across Two Orbital Regimes

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Key Points:

- A flux-free finite-state trigger detects solar proton onsets onboard from raw counts using a rolling robust median plus MAD background
- The causal rolling trigger matches an oracle fixed threshold tuned with full hindsight so no multi-year archive is needed and cold start is days
- The identical engine transfers to two low Earth orbit monitors and the channel best by static statistics is not the one that triggers best

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Abstract

Small satellites need to detect solar energetic particle (SEP) events onboard, in real time, from raw counts—cheaply enough to run continuously as a trigger for heavier inference. Existing onset methods assume calibrated flux or run offline. We test an instrument-agnostic, finite-state trigger that works directly on Level-0 counts using a causal rolling robust background (median plus a multiple of the median absolute deviation). On 5.7 years of geostationary KSEM/GK2A counts, the proton trigger matches a fixed threshold tuned with full hindsight (POD 0.59, FAR 0.11), needs no multi-year archive, and becomes reliable within days. It fails for electrons, whose enhancements rise too gradually. Transferred unchanged to two LEO monitors (POES MetOp-03 and NOAA-19), it detects proton onsets on both—and the channel best by static statistics is not the one that triggers best. The result is a flux-free, cold-start-tolerant onset trigger, validated across two orbital regimes, as a rule-based layer for gating onboard prediction.

1 Introduction

1.1 Radiation hazards in low Earth orbit and the need for autonomy

Small satellites in low Earth orbit (LEO) are continuously exposed to a space radiation environment that is both hazardous and difficult to predict. Energetic particles produce total ionizing dose effects and single event effects (SEE) in spacecraft electronics, of which single event latch-up (SEL) can destroy a device within seconds (Jun et al., 2024). Several CubeSat missions have suffered radiation-induced early termination: IDEASSat lost power-system function to an SEL roughly three weeks after launch (Chiu et al., 2022), and comparable failures have been attributed to inadequate latch-up protection (Bouwmeester et al., 2017) and to radiation damage of boot memory (Langer et al., 2015). The LEO environment is especially variable because of the South Atlantic Anomaly, where trapped-particle exposure rises by orders of magnitude during each pass, and because solar energetic particle (SEP) events reach LEO within tens of minutes of a solar eruption, with little warning. The ISS safety framework treats this environment as an actively managed hazard rather than a static background (National Aeronautics and Space Administration, 2019).

Mitigating these hazards in real time is in tension with how CubeSats are operated. The prevailing paradigm is ground-dependent: data are collected, downlinked, analyzed on the ground, and commands are uplinked. UHF rates of order 9.6 kbit/s and a handful of short ground contacts per day mean a spacecraft is unsupervised for the large majority of each day (Chiu et al., 2022; Cichy et al., 2005), during which a latch-up that unfolds in seconds cannot wait for a ground decision. The need is more acute for crewed deep-space exploration, where one-way light time (minutes to tens of minutes to Mars) makes ground-commanded evacuation impossible within the tens-of-minutes golden time before an SEP dose accumulates (Zaman et al., 2021; Löwe et al., 2025). On-board autonomous detection and response is therefore a survival requirement rather than a convenience.

1.2 On-board autonomy: the state of the art and what is missing

On-board autonomy for Earth-observation and telemetry tasks has advanced rapidly, but it has concentrated on tasks other than pre-emptive radiation monitoring. Recent missions optimize heavy 2-D optical inference (Phi-sat-2) (European Space Agency, 2024) or post-hoc telemetry anomaly detection (OPS-SAT) (Ruszczak et al., 2025), both of which act on data that already exist rather than issuing a pre-emptive alert before a hazard develops. National roadmaps now identify lightweight, 1-D time-series, on-board pre-emptive inference as a target capability (Korea AeroSpace Administration, 2025), yet it has not been demonstrated in orbit.

A parallel literature forecasts SEP events with machine learning, but almost entirely from ground-based, multi-source inputs: solar flare, CME, and radio burst features (Ali et al., 2025), or SHARP active-region magnetogram parameters (Kasapis et al., 2024), typically as models that run after the fact on the ground rather than continuously on board. Lightweight 1-D particle-time-series models have been argued to be feasible for on-board use (Stumpo et al., 2024; Torres et al., 2025; Yi et al., 2025), but the trigger that would gate such a model on board, deciding cheaply and continuously when the expensive inference should run, has not itself been characterized on operational data.

This gap motivates the present work. In the PADO-SAT CubeSat concept, the on-board pipeline is event-driven: a rule-based low-power trigger monitors the particle stream continuously and, only when it fires, wakes the heavier prediction model under the supervision of the on-board computer. This paper evaluates that trigger layer. It is not a forecasting model and does not attempt full SEP prediction; it is the cheap, always-on front end whose job is to separate candidate events from background reliably enough to gate what follows.

1.3 Why raw counts, and the research gap

Existing automated SEP onset methods fall into two families, and neither is designed for the trigger role just described. Statistical change-point methods based on the Poisson-CUSUM scheme are well established for onset determination (Huttunen-Heikinmaa et al., 2005) and remain in active use, including in hybrid bootstrap form for onset-time uncertainty (Palmroos et al., 2025) and in recent multi-spacecraft catalogs (the SERPENTINE catalog) (Dresing et al., 2024). These methods are statistically principled but operate on calibrated flux and are applied as post-hoc, ground-based analysis tools; their authors note that a noisy, slowly drifting, or poorly estimated background can produce false or premature detections, which is precisely the regime an on-board trigger must survive. Fixed-background methods, in which an event is declared when a calibrated flux exceeds a preset level (Papaioannou et al., 2014; Moreland et al., 2024), are operationally simple but again presuppose flux.

The one close structural precedent for count-based detection is the AMS-02 real-time SEP monitor on the ISS, which detects events from increases in the instrument trigger rate rather than calibrated flux (Faldi et al., 2023). AMS-02, however, classifies background by McIlwain L -parameter, an approach that presupposes both the high count statistics of a large magnetic spectrometer and the on-board computation of magnetic coordinates; neither is available to a low-statistics, resource-constrained CubeSat, so the method does not transfer directly.

The most direct methodological precedent for a count- and rate-based trigger is the nowcasting system of Löwe et al. (2025) for the MSL/RAD dosimeter, which signals an event when the dose rate exceeds its slowly varying background by a fixed fraction of 25%. That fixed-fraction rule is well matched to RAD: a high-statistics dosimeter whose single dose-rate channel has a smooth, slowly varying background, with a coefficient of variation of order a few percent in deep space. CubeSat-scale logic-channel counts are the opposite regime—low in statistics, strongly overdispersed, and spread across channels whose quiet backgrounds differ by orders of magnitude—so a single fixed fractional threshold cannot serve all channels simultaneously, and a fixed offset cannot track the per-channel, time-varying background. The gap is therefore not merely the absence of raw-count processing but the absence of an estimator that adapts its scale to the per-channel, time-varying background each cadence. This motivates replacing the fixed fraction with a per-channel rolling median and the implicit scale assumption with a robust median absolute deviation (MAD); the ablation of Section 4 shows that the robust scale is in fact the single most consequential design choice.

To our knowledge, no existing method provides at once a trigger that is simultaneously (i) flux-free, operating on Level-0 raw counts without instrument-specific calibration or energy-band assumptions; (ii) causal and cold-start-tolerant, usable within days of deployment rather than requiring a multi-year archive; and (iii) channel-agnostic, applying the same logic across all detector channels so that channel suitability, and channel failure, emerge from the method itself. The contribution of this paper is to assemble rolling robust-background estimation, catalog-based evaluation, and per-channel diagnostics into a single trigger and to validate it end-to-end on multi-year geostationary particle-detector data, quantifying—to our knowledge for the first time—how this combination behaves in the CubeSat trigger setting. The individual ingredients are standard; their integration and operational characterization for this purpose are not.

1.4 Why static separability is not trigger suitability, and contributions

Underlying the choice of raw counts is a distinction that this paper makes explicit and then quantifies: for a trigger, the relevant property of a channel is how cleanly its count distribution separates events from background under the actual detector, which need not coincide with how clean that distribution looks under static, whole-record statistics. A channel can appear distributionally noisy yet trigger well, or appear clean yet trigger poorly. We show that the channel a static reading of the counts would prefer is not the channel that performs best as a dynamic trigger, and that the best trigger channels differ between protons and electrons.

The specific contributions are:

1. A flux-free, instrument-agnostic SEP onset trigger built on a rolling robust (median + MAD) background, evaluated against the NOAA proton and SWPC electron catalogs across all KSEM/GK2A channels.
2. A demonstration that the causal rolling trigger matches an “oracle” fixed-threshold baseline—a fixed threshold tuned with full hindsight over the entire multi-year record, and therefore an upper bound that no real-time system could attain—establishing that no pre-accumulated archive is needed for on-board use, and that the cold-start interval is days rather than months.
3. A diagnosis of channel failure modes, including the automatic identification of a hardware-noisy channel, that provides a concrete criterion for on-board channel selection; together with an honest negative result for electrons, where the threshold-crossing principle is shown to be physically mismatched to gradual enhancements.
4. A transfer of the identical trigger to two LEO detectors (POES MetOp-03 and NOAA-19), demonstrating reproducible proton onset detection in a second orbit regime and showing that the channel favoured by static count statistics differs from the channel that performs best as a dynamic trigger—the LEO counterpart of the static-vs-dynamic distinction.

The remainder of the paper describes the data and instruments (Section 2), the detection framework and evaluation (Section 3), the results for the geostationary development instrument followed by the LEO validation (Section 4), and the operational implications and limitations (Section 5), before concluding (Section 6).

2 Data and Instrumentation

2.1 KSEM particle detector aboard GK2A

The particle count data used in this study come from the Particle Detector (PD) of the Korea Space WEather Monitor (KSEM), a space-weather payload on the geostationary satellite GEO-KOMPSAT-2A (GK2A), positioned at 128.2 deg E longitude in

an equatorial geostationary orbit at a geocentric distance of about 6.6 Earth radii ($\sim 35,800$ km altitude) (Seon et al., 2020). The PD inherits the design heritage of the THEMIS Solid State Telescope and consists of three sensor modules (PD1, PD2, PD3) viewing different directions relative to the spacecraft; each module houses two telescopes mounted back-to-back, so that the system collects data on six physical telescope sides in total (three modules times two sides). The silicon detectors use foils and magnets to separate electrons from ions, and the instrument measures electrons over roughly 0.1–3.8 MeV and ions over roughly 0.15–22.5 MeV with a one-second effective cadence (Seon et al., 2020).

Particle identification is performed in hardware by coincidence logic among the silicon detector elements. The charge-signal combination produced as a particle traverses the stacked detectors, expressed through the detector elements O, F, T, and U, is mapped to the valid physical count channels used here: the proton-side channels O, OU, and OUT, the electron-side channels F, FT, FTU, and FTUO, and a cosmic-ray (CR) channel (Seon et al., 2020; Lee, 2018). We use these on-board logic-channel counts, the Level-0 product, rather than the calibrated differential flux, because counts are the quantity available on board without calibration, consistent with the trigger role defined in Section 1.

The 6 MeV response that recurs in the analysis has a specific hardware meaning. The official KSEM proton energy classification (Seon et al., 2020) divides the proton range into P1 (0.1–0.4 MeV), P2 (0.4–2.0 MeV), P3 (2.0–6.0 MeV), P4 (6.0–15.0 MeV), and P5 (15.0–30.0 MeV). The 6 MeV value is thus the hardware boundary between the P3 and P4 proton channels, and the logic combinations that constitute the proton trigger channels (OU, OUT) respond across this boundary, sensing the higher-energy proton flux as it crosses from P3 into P4. This 6 MeV integral response does not coincide with the >10 MeV band of the NOAA proton catalog used as ground truth (Section 2.3); we therefore evaluate channel behavior against the catalogs directly rather than through a flux proxy, as described in Section 3.6.

2.2 Count series and preprocessing

The count record spans 2019-05-01 to 2024-12-31, covering the rising phase and maximum of solar cycle 25. The native one-minute series is resampled to a uniform 15-minute cadence by averaging, which suppresses single-sample noise while preserving the onset timing relevant to an SEP rise of tens of minutes to hours; this 15-minute cadence matches the operational data product used for RAD-based SEP nowcasting by Löwe et al. (2025). The resulting L0 input is a 54-dimensional vector per time step: the six telescope sides each contribute the eight coincidence-logic count channels plus one instrument monitoring/overflow channel that records status, calibration flags, or out-of-range counts (6 sides times 9 channels = 54). After resampling, each channel spans on the order of 2×10^5 time steps—i.e. the number of 15-minute samples over the 5.7-year record—giving a working array of 54 channels $\times \sim 2 \times 10^5$ samples. Each channel is processed independently, and the analysis of trigger performance focuses on the physical coincidence-logic channels rather than the monitoring channels.

2.3 Ground-truth event catalogs

Detections are evaluated against two independent event catalogs.

For protons we use the NOAA GOES solar proton event (SPE) list, defined on the >10 MeV integral proton channel with the standard event threshold of 10 pfu. Restricted to the KSEM data span and this threshold, the catalog provides 34 proton events.

For electrons we use the SWPC ALTEF3 >2 MeV electron alert list, with an event threshold of 1000 pfu. Over the same span this provides 119 electron events.

The event counts quoted above (34 proton, 119 electron) are those falling within the KSEM span of Section 2.2; the LEO validation of Section 2.5 uses the same two catalogs but over a slightly longer span (2019–2025), which admits 42 proton and 159 electron events. We report each count against the data span it is matched to, rather than a single global figure.

Each catalog entry supplies an event begin (onset) time and a peak time, which are the reference points for the matching and profiling procedures of Section 3. We note two caveats that the evaluation must tolerate. First, the catalog onset and peak times are themselves uncertain, which motivates the generous ± 24 h matching tolerance adopted in Section 3.5. Second, the electron catalog peak corresponds to a daily-maximum reporting convention that is coarser than the proton onset definition; this is one reason the electron evaluation is treated separately and interpreted conservatively in Section 4.3.

2.4 Channel grouping by quiet-background behavior

For the analysis it is useful to group the logic channels by how they behave during quiet intervals, since this directly affects their suitability as triggers. We distinguish three groups. Group A (OU, OUT) has a very low and stable quiet background. Group B (FTU, FTUO) has an intermediate quiet background. Group C (O, F, FT, CR) has an active, high, or strongly varying quiet background. These labels describe observed background behavior only; they are not assumed to predict trigger suitability, which is established empirically in Section 4 and which, as the results show, does not follow the static distributional ranking of the channels. Quantitatively, the median count during proton events exceeds the quiet-interval median by a factor of about 7–14 for Group A (OU/OUT), 2.7–3.9 for Group B (FTU/FTUO), and below 2.5 for Group C (O/F/FT/CR), confirming the group ordering used here.

2.5 POES/MetOp SEM-2 particle detectors (LEO validation)

To test whether the trigger developed on KSEM transfers to the low-Earth-orbit (LEO) environment of the PADO-SAT target, we apply the identical detection engine (Section 3.8) to two further detectors: the Medium Energy Proton and Electron Detector (MEPED) of the Space Environment Monitor-2 (SEM-2) package aboard MetOp-03 and NOAA-19. Both are Sun-synchronous polar orbiters at an altitude of about 850 km with an orbital period of about 100 min (Evans & Greer, 2000; Sandanger et al., 2015). The two spacecraft differ in age: NOAA-19 was launched in 2009 and MetOp-03 in 2018, a difference that matters because MEPED proton detectors degrade with accumulated fluence and require correction two to three years after launch, the 90-degree telescope degrading faster than the 0-degree one (Sandanger et al., 2015). This degradation is the physical basis for an isolated, anomalous mid-latitude count excursion—a spurious count spike not associated with any real flux change—that we later identify on NOAA-19 (Section 4.7).

MEPED carries two solid-state proton telescopes, the 0-degree (zenith-viewing) and 90-degree (anti-velocity) detectors, each with a 30-degree full-width field of view, together with an omnidirectional detector system. The directional proton channels span P1 through P6 with nominal thresholds 30–80, 80–240, 240–800, 800–2,500, 2,500–6,900 keV, and >6,900 keV (Evans & Greer, 2000); the electron channels E1–E3 cover >43, >114, and >292 keV after optimised calibration (Ødegård et al., 2017); and the omnidirectional channels P6–P9 register protons above roughly 16 MeV. We use the on-board count rates of these channels as the Level-0 product, exactly as for KSEM. Each channel is treated as one count series, indexed by (species, direction, energy), giving 22 physical channels per spacecraft; the count caches are stored per channel, with the sub-satellite geographic position (latitude, longitude, altitude) held separately for the South Atlantic Anomaly tagging of Section 3.8.

Table 1. The three particle monitors used in this study. KSEM (GEO) is the development detector; the two POES/MetOp SEM-2 detectors (LEO) are the validation detectors.

Property	KSEM / GK2A (GEO)	POES MetOp-03 (LEO)	POES NOAA-19 (LEO)
Orbit	Geostationary, 128.2 deg E, $\sim 6.6 R_E$	Sun-sync polar, ~ 850 km	Sun-sync polar, ~ 850 km
Period	24 h (fixed longitude)	~ 100 min	~ 100 min
SAA transits	None	Yes (counts rise by orders of magnitude)	Yes
High-latitude precipitation	None	Yes	Yes
Proton channels	Logic O/OU/OUT (6 MeV P3/P4 boundary)	Directional P1–P6, omni P6–P9 (>16 MeV)	Same
Look directions	6 telescope sides	0 deg / 90 deg telescopes	Same
Launch / age	2018 (GK2A)	2018	2009 (older; degradation)
Trigger preprocessing	Channel selection only	Channel selection + SAA tagging	Same, plus aged-detector artefact
Reference	Seon et al. (2020)	Evans and Greer (2000)	Evans and Greer (2000)

Two known MEPED caveats bear directly on the channel behaviour reported in Section 4.7. First, in the absence of high-energy solar protons in the P5 channel, the directional P6 channel is contaminated by relativistic ($\gtrsim 756$ keV) electrons (Yando et al., 2011), so a P6 count is not unambiguously a proton signal. Second, the omnidirectional channels respond to the entire sky and therefore mix solar-proton signal with the locally trapped population encountered in the South Atlantic Anomaly and at high latitudes; whether this contamination actually degrades a channel’s trigger performance is, as the results show, a question that static count statistics and the dynamic trigger answer differently (Section 4.7, Section 5.4).

The KSEM logic-channel grouping of Section 2.4 has a direct counterpart here. The low, stable quiet background of KSEM Group A (the OU/OUT proton-trigger channels) corresponds on POES to the higher-energy directional proton channels (notably `pro_tel0 P4/P5`) and the omnidirectional proton channels, which are the LEO channels we examine as proton triggers; the active, always-on KSEM Group C corresponds to the low-energy POES channels that carry signal continuously and are correspondingly poor triggers.

2.6 Orbit and detector environment across the three monitors

The three monitors span the two orbital regimes relevant to the mission and differ in exactly the properties that the trigger must tolerate. The comparison is summarised in Table 1.

The central methodological point of the orbit comparison is that the same count-based trigger requires no preprocessing beyond channel selection on the clean geostationary background, but on LEO must additionally contend with the periodic background structure of SAA transits and high-latitude precipitation; how the trigger is adapted to

that structure without sacrificing detection is the subject of Section 3.8, and how the adaptation behaves is the subject of Section 4.7.

3 Methods

This section describes the detection framework used to identify solar energetic particle (SEP) onsets directly from raw particle count data. We first state the design philosophy (Section 3.1), then specify the common detection engine shared by all variants (Section 3.2), the proposed rolling robust-background estimator (Section 3.3), the fixed-threshold baseline that serves as the ablation control (Section 3.4), the catalog-based evaluation protocol (Section 3.5), and the event count profiling procedure used to determine channel-specific thresholds (Section 3.6). We then describe how the detector is realized as an on-board flight-software architecture (Section 3.7), and finally how the identical engine is transferred to the two LEO detectors for validation, including the South Atlantic Anomaly handling that the LEO environment requires (Section 3.8).

3.1 Design philosophy

The detector is intended to run on board a resource-constrained CubeSat as the low-power, rule-based trigger layer that precedes any heavier on-board inference. Three constraints follow from this setting. First, the trigger must operate on Level-0 (L0) raw counts: converting counts to a calibrated particle flux requires instrument-specific calibration, energy-band assumptions, and cross-channel corrections that add computational burden and latency, and that presuppose ground processing the spacecraft cannot perform autonomously in real time. Second, the trigger must be available immediately after deployment, without waiting for a multi-year archive from which to derive operating points. Third, the trigger must be instrument-agnostic in form: the same logic and the same hyperparameters (window length W , multiplier k , persistence count, `peak.floor`) apply unchanged across all channels and both detectors, and the only quantities that differ between channels are the background median and robust scale, which the estimator computes automatically from each channel’s own trailing counts. No per-channel hand-tuning is performed.

These constraints motivate a central methodological position of this work, which we state here and quantify in Section 4: for a trigger, *a channel’s static count separability is not the same as its dynamic trigger suitability*. The quantity that matters is not how clean a channel’s count distribution looks under static, whole-record statistics, but how cleanly it separates event intervals from the quiet background once that channel drives the rolling-background detector. A channel may look distributionally noisy yet trigger well, or look clean yet trigger poorly.

3.2 Common detection engine

All detector variants share the same detection back end and differ only in how the threshold is computed. Raw counts are first resampled to a uniform 15-minute cadence by averaging, which suppresses single-sample noise while preserving onset timing at the resolution relevant to a multi-tens-of-minutes SEP rise. All three photodiodes (PD1, PD2, PD3), both detector sides (A, B), and all eight logic channels are processed independently.

For a channel count series $c(t)$ and a threshold series $T(t)$, a candidate event is a maximal run of consecutive samples satisfying $c(t) \geq T(t)$ whose duration is at least $D_{\min} = 1$ h. For each candidate run the onset time is the first sample of the run, the peak time is the sample of maximum count within the run, and the end time is the last sample. A candidate is retained as a detected event only if its peak count is at least a post-filter floor, `peak.floor` (default 2 counts); this rejects marginal runs that cross the threshold without reaching a physically meaningful amplitude. The 1-hour persistence

requirement and the `peak_floor` filter are identical across all variants, so that any performance difference between variants is attributable solely to the threshold definition.

The evaluation in Section 4 applies this engine as an offline batch over the full count record, which is the appropriate form for catalog-referenced scoring because the `peak_floor` post-filter inspects the maximum count over a completed candidate run. The on-board realization cannot look ahead to a run’s peak, so it performs the same decision causally, one sample at a time, replacing the run-length and `peak_floor` filters with a consecutive-sample persistence check; the two forms are equivalent in steady state for the abrupt proton onsets the trigger targets. The offline form is used throughout Section 4 so that the reported POD/FAR are directly comparable to the catalogs, while the causal on-board form, and its cost, are described in Section 3.7 and measured in Section 5.1.

3.3 Proposed method: rolling robust background

The proposed trigger estimates a time-varying threshold from a trailing window, so that it adapts to seasonal and orbital drift in the background and requires no future information. The background is recomputed once per day from the preceding W days of counts, with $W = 30$ d as the default. The once-per-day update follows the daily-updated trailing-background approach of Löwe et al. (2025), who fit the background over the preceding five days; the 30-day window is our own default, and the window sweep of Section 4 (Table 6) shows that performance is flat from 5 to 30 days, bracketing and remaining consistent with the shorter Löwe window. Within each window the background level is summarized by its median, m , and its scale by a robust estimate of dispersion,

$$\sigma = 1.4826 \cdot \text{median}(|x - \text{median}(x)|), \quad (1)$$

the median absolute deviation (MAD) rescaled to be consistent with the standard deviation under normality. The threshold is then

$$T(t) = m + k \sigma, \quad (2)$$

with $k = 10$ by default. The estimator is forward-filled until the next daily update, so that the threshold at any time depends only on past data, consistent with causal on-board operation.

Two design choices distinguish this estimator. The first is the use of MAD rather than the sample standard deviation. The KSEM count channels are strongly overdispersed relative to a Poisson expectation, so a single spike contaminating the window inflates the standard deviation and destabilizes the threshold; the MAD is insensitive to such outliers and keeps the threshold stable for the low-count channels on which proton detection relies. The proton-trigger channels (OU/OUT) have a standard-deviation-to-MAD ratio of roughly 1,500–5,000, so the sample standard deviation is set almost entirely by rare spikes rather than by the quiet background and cannot serve as a dispersion estimate; this is the quantitative basis for the MAD-to-standard-deviation collapse reported in Section 4 (POD 0.591 to 0.259). The second is the optional restriction of the background sample to the quietest days within the window: we compare using the entire window (denoted `quietoff`) against retaining only the seven days of lowest daily median (`quiet7`), the latter a variant of our own, loosely inspired by the practice of excluding event-contaminated samples from the background estimate. These two axes (MAD vs. standard deviation, full window vs. quietest-seven) define the four rolling variants used in the ablation of Section 4.

The contrast with the closest count-based precedent clarifies why an adaptive, robust estimator is required rather than a simpler rule. The MSL/RAD nowcasting sys-

tem of Löwe et al. (2025) triggers when the dose rate exceeds a slowly varying background by a fixed fraction (25%). That fixed-fraction rule is well matched to its instrument: RAD is a high-statistics dosimeter whose single dose-rate channel has a smooth, slowly varying background, with a coefficient of variation of order a few percent in deep space. CubeSat-scale logic-channel counts are the opposite regime, low in statistics, overdispersed, and spread across channels whose quiet backgrounds differ by orders of magnitude, so a single fractional threshold cannot serve all channels and a fixed offset cannot track the per-channel, time-varying background. Replacing the fixed fraction with a per-channel rolling median, and the implicit Gaussian-scale assumption with a robust MAD scale, is therefore not a refinement of the Löwe rule but the adaptation its instrument did not require; the ablation of Section 4 shows that the robust scale is in fact the single most consequential of these choices. The use of a rolling average as an on-board noise-reduction step is not itself new: the next-generation CubeSat detector REPTile-2 on CIRBE applies a rolling-sum averaging stage in its FPGA firmware to suppress high-frequency noise before peak detection (Li et al., 2022), a precedent for trailing-window smoothing in resource-constrained on-board processing. What differs here is the purpose and the statistic: a robust median/MAD background estimator for event triggering rather than a sum-averaging filter for pulse cleaning.

3.4 Fixed-threshold oracle baseline

To isolate the contribution of the rolling estimator itself, we define a fixed-threshold *oracle* baseline that removes background estimation entirely. The detection back end of Section 3.2 is retained unchanged, but the threshold is a single time-invariant constant per channel,

$$T(t) = C, \tag{3}$$

with no rolling update and no quiet-day selection. The baseline blends no floor into the threshold, keeping the constant pure so that the ablation varies a single factor at a time.

We consider two ways of setting C , corresponding to two established fixed-background conventions. In the *constant* mode, C is assigned per channel (or per logic type) as an absolute count level, following the operator-specified quiet-background practice of Papaioannou et al. (2014). In the *percentile* mode, C is the 95th percentile of the channel’s entire count record. The choice of the 95th percentile is empirical: across the proton trigger channels (OU, OUT) and the electron trigger channels (FTU), the quiet 95th percentile coincides closely with the separation-optimal count threshold of Section 3.6, so the two criteria converge at p95.

A key property of the oracle baseline must be stated explicitly, because it governs the interpretation of the results. The constant C , whether absolute or percentile, is computed from the full multi-year record. It therefore uses information that would not be available to a spacecraft operating in real time, and in particular not available shortly after launch. The oracle baseline thus represents the best a static threshold could do with hindsight over the entire mission, an upper bound rather than an operationally achievable system. The rolling method of Section 3.3, by contrast, uses only the trailing 30 days (or fewer; see the window comparison in Section 4) and is causal. Any finding that the rolling method merely matches the oracle is therefore a finding in favour of the rolling method, since it attains comparable performance under strictly harder, operationally realistic conditions.

3.5 Catalog-based evaluation

Detected events are evaluated against two ground-truth catalogs: the NOAA GOES solar proton event list (>10 MeV protons, events of at least 10 pfu) and the SWPC AL-TEF3 greater-than-2-MeV electron alert list (events of at least 1000 pfu). Over the KSEM data span (2019–2024) these yield 34 proton events and 119 electron events, respectively. The LEO validation of Section 3.8 applies the same two catalogs over the slightly longer POES span (2019–2025), which admits 42 proton and 159 electron events; we score each detector against the catalog counts of its own matched span rather than a single global figure.

A detected event is counted as a hit if its onset time falls within a tolerance $\text{tol} = 24$ h of a catalog begin time; the tolerance is set generously because catalog onset and end times are themselves uncertain and because the count threshold introduces a floor-dependent onset delay. From the hit assignment we compute the probability of detection,

$$\text{POD} = \frac{\text{number of catalog events matched}}{\text{number of catalog events}}, \quad (4)$$

and the false alarm ratio,

$$\text{FAR} = \frac{\text{number of detections not matched to any catalog event}}{\text{number of detections}}. \quad (5)$$

We additionally report POD stratified by event magnitude (10–100, 100–1000, and >1000 pfu) to separate the detection of weak from strong events. For matched pairs we record the onset time difference (KSEM onset minus catalog begin); a negative value indicates that the KSEM count crossing leads the catalog onset. Where a single balanced figure is useful for ranking channels, we report the difference $J' = \text{POD} - \text{FAR}$.¹

3.6 Event count profiling for threshold determination

Determining a defensible constant C for the oracle baseline, and more generally understanding how each channel behaves during events, requires characterizing the count distribution inside event intervals relative to the quiet background. We do this directly from the catalogs rather than through a flux proxy, which avoids the energy-band mismatch that would otherwise contaminate the proton channels: the OU and OUT logic channels have a 6 MeV cutoff that prevents validation against a flux proxy, so a catalog-referenced profile is the only unbiased way to observe their event behavior.

For each channel we define an event mask from the catalog intervals and a quiet mask from their complement. A wide event mask spanning onset to peak + 72 h (proton) or peak + 48 h (electron) captures the full event but dilutes the signal, because the long post-peak decay and pre-rise interval contribute low counts; we therefore also compute a narrow mask spanning onset to peak + 12 h and a per-event peak level, defined as the maximum count within ± 6 h of each catalog peak. Separation statistics are computed on the narrow mask to avoid dilution.

From these distributions we derive two threshold candidates per channel. The first is a quiet-percentile threshold (the 50th through 99.9th percentiles of the quiet counts),

¹ $J' = \text{POD} - \text{FAR}$. At the event level the number of true negatives is not well defined, so the false-alarm ratio FAR (false alarms per detection) is used in place of a false-positive rate; J' is therefore a Youden-like balanced score rather than the Youden statistic itself. Only POD and FAR are computed directly; J' is reported as their difference.

following the stationary-background convention of Papaioannou et al. (2014). The second is a separation-optimal threshold obtained without external libraries: over the union of event and quiet count values we select the count value that maximizes the per-channel separation score $J' = \text{POD} - \text{FAR}$. As noted in Section 3.4, the quiet 95th percentile coincides closely with this separation-optimal threshold for the trigger channels, which is why p95 is adopted for the percentile mode of the oracle baseline.

3.7 On-board realization as a cFS-style flight architecture

The detection engine of Sections 3.2–3.3 specifies *what* the trigger computes. This section specifies *how* it is realized as on-board flight software, because the contribution of the trigger is not the detection rule in isolation but a layer that can run autonomously on a spacecraft. We implement the trigger following the application structure of NASA’s Core Flight System (cFS), the flight-software framework whose layered design is intended to transfer across processors and missions, and we evaluate the realization on the edge platform of Section 5.1. Mapping the trigger onto a recognized flight architecture, rather than an ad hoc script, is what makes the on-board claim concrete and is the bridge from the offline evaluation of Section 4 to the autonomous operation that motivates the mission.

3.7.0.1 Decomposition into three applications. The cFS Limit Checker (LC) and Stored Command (SC) applications provide an exact template for an event-driven monitor-and-respond layer, and we adopt their division of labor. The trigger is realized as three communicating nodes connected only through a publish/subscribe message bus (the cFS Software Bus role, here served by the ROS 2 transport), so that the nodes share no direct state and can be added or replaced independently:

- a *watchpoint* node (WP; cFS LC watchpoint role), which holds the rolling background and emits, once per cadence and per monitored channel, a three-valued watch result (TRUE/FALSE, with STALE and ERROR for missing or invalid data);
- an *actionpoint* node (AP; cFS LC actionpoint role), which combines the per-channel watch results into a single operational state and decides when an alert is warranted;
- a *stored-command* node (SC; cFS SC role), which receives the alert and issues the corresponding mode-transition command sequence.

This is the same monitor (LC) to response (SC) chain that cFS missions use for autonomous fault response; here the monitored quantity is the SEP count trigger rather than a house-keeping limit.

3.7.0.2 Multi-channel combination by RPN evaluation. The AP combines channels using the LC reverse-Polish (RPN) actionpoint mechanism, in which a fixed postfix logic equation over the channel watch results is evaluated each cadence. The default equation is the disjunction of the monitored OU/OUT channels, so that an onset on any suitable channel raises the candidate condition, consistent with the channel-selection result of Section 4.1; the same mechanism supports conjunction, exclusive-or, and negation, so that a mission can encode a more selective voting rule (for example, requiring agreement among channels) by changing only the equation, with no change to the engine. The evaluation is three-valued: STALE and ERROR watch results propagate through the logic rather than being silently treated as FALSE, so that a data gap on one channel cannot masquerade as a quiet background and cannot by itself either force or suppress an alert.

3.7.0.3 Persistence and latching. A single cadence in which the equation evaluates TRUE does not raise an alert. Following the LC actionpoint persistence count, the condition must hold for N_p consecutive cadences ($N_p = 3$ by default, spanning 45 min at the 15-minute cadence) before the AP promotes the state to an alert; any interven-

ing FALSE resets the count, and a STALE or ERROR result also resets it rather than advancing toward an alert. This consecutive-sample persistence is the causal counterpart of the offline run-length and `peak_floor` filters of Section 3.2: it suppresses isolated single-cadence excursions without looking ahead to a completed event, which the offline `peak_floor` cannot do on board. Once the alert is raised the AP latches and stops re-issuing the trigger until the condition clears, matching the LC convention of setting an actionpoint passive after it fires to prevent repeated command bursts.

3.7.0.4 Configurability without code change. The window length W , the threshold multiplier k , the persistence count N_p , and the RPN equation are all exposed as run-time parameters rather than compiled constants, following the cFS table-driven configuration philosophy in which operating points are adjusted by uploading a parameter table rather than rebuilding the flight software. This is what allows the window-length and threshold trade-offs characterized in Sections 4.4–4.5 to be exercised operationally, and it is the same property that makes the trigger instrument-agnostic in the sense of Section 3.1: porting to a different detector changes only the input adapter and, optionally, the RPN equation, while the engine and its hyperparameters are unchanged.

The on-board cost of this realization—per-update inference time, latency through the three-node chain, and power—is measured directly and is reported in Section 5.1, where it quantifies the claim that the layer is light enough to run continuously as a gate.

3.8 Transfer to LEO detectors and South Atlantic Anomaly handling

The instrument-agnostic claim of Section 3.1 is tested by applying the detector to the two POES/MetOp SEM-2 instruments of Section 2.5. The transfer is deliberately minimal, so that any change in behaviour is attributable to the detector and orbit rather than to the algorithm.

3.8.0.1 The detection engine is unchanged. The five functions that constitute the engine—the robust scale of Equation (1), the quiet-sample selection, the rolling background, the threshold of Equation (2), and the segment detector of Section 3.2—are byte-for-byte identical between the KSEM and POES implementations; we verified this by abstract-syntax-tree comparison (zero diff). The operating points are also carried over unchanged ($W = 30$ d, $k = 10$, onset floor 0.5, `peak_floor` 2). What differs is only the input/output adapter: the POES reader iterates over the 22 (species, direction, energy) channels of Section 2.5 in place of the KSEM logic channels, and the catalog evaluation of Section 3.5 is applied identically. This makes the LEO study a true transfer of the same trigger, not a re-tuned reimplementation.

3.8.0.2 The South Atlantic Anomaly is handled by post-hoc tagging, not by masking. The one feature the geostationary development environment did not present is the periodic count enhancement of SAA transits and high-latitude precipitation (Section 2.6). A naive response would be to mask the SAA region out of the input before detection, but this discards real observing time and, more importantly, would remove any genuine SEP onset that happened to begin during a transit, depressing the true-positive rate. This concern is well founded for the POES orbit: on the structurally similar LEO REP-Tile instrument, the detector is exposed to trapped inner-belt protons only while passing through the SAA region (longitude roughly 0–60 degrees), where the weaker local field lets particles mirror at lower altitude so that more reach the spacecraft (Li et al., 2022). The SAA is therefore a localized, longitude-bounded enhancement rather than a whole-orbit one, which is exactly why whole-interval masking would discard far more good observing time than it needs to. We therefore run the detector over the full count series exactly as for KSEM, and only after an onset is detected do we look up the geomagnetic conditions at the onset sample. The geomagnetic field magnitude $|B|$ is computed from the International Geomagnetic Reference Field (IGRF-13/14; Alken et al.

Table 2. Channel discrimination by separation score J' on the narrow event mask. Higher J' means cleaner event-versus-quiet separation. The separable group is reversed between protons (Group A) and electrons (Group B/FT).

Group	Proton J' (mean)	Proton J' (max)	Electron J' (mean)	Electron J' (max)
A (OU/OUT)	0.716	0.835	0.176	0.528
B (FTU/FTUO)	0.180	0.261	0.663	0.707
C (O/F/FT/CR)	0.055	0.154	0.506	0.711

(2021), evaluated with the updated 2024 coefficients through the ppigrf implementation) at the sub-satellite position, and an onset is flagged as SAA-associated when $|B| < 25,000$ nT, the conventional field-strength contour for the anomaly; the onset's $|B|$ and magnetic latitude are recorded alongside the detection. This post-hoc tagging preserves the true-positive rate in full—no onset is ever suppressed—while still allowing SAA-driven false alarms to be identified and quantified separately, through the per-channel counts `n_fa_saa` (false alarms whose onset is SAA-associated) and its fraction `n_fa_saa_frac` reported in Section 4.7. Whether SAA contamination actually reaches the detected onsets, or is instead absorbed into the rolling background before it can trigger, is then an empirical question the tagging lets us answer rather than presuppose.

This separation—identical engine, identical operating points, LEO-specific diagnostics applied only after detection—is what lets Section 4.7 read as a clean test of transfer: the proton-onset performance on two LEO detectors is produced by exactly the trigger characterised on GEO, with the orbit's structured background made visible through tagging rather than hidden by preprocessing.

4 Results

4.1 Channel discrimination and selection

The suitability of each logic channel as a trigger is established empirically rather than assumed from its quiet-background group. Table 2 reports the event-versus-quiet separation of each channel group, measured by the separation score J' on the narrow event mask, against each catalog. The separable group is reversed between species: protons separate best on Group A (OU/OUT), whereas electrons separate best on Group B and the FT channels. This reversal is the first concrete instance of the static-versus-dynamic distinction of Section 3.1 and motivates evaluating proton and electron triggers on different channels.

4.2 Proton trigger: rolling background versus the oracle baseline

On the proton-trigger channels (OU/OUT), the causal rolling trigger ($W = 30$ d, $k = 10$) matches the oracle fixed-threshold baseline that was tuned with full multi-year hindsight. Table 3 reports per-channel POD/FAR against the NOAA proton catalog, together with the median onset lead. Averaged over the OU/OUT channels (excluding the noise channel identified below), the rolling trigger reaches POD 0.591 at FAR 0.112, against 0.586/0.136 for the percentile-mode oracle and 0.586/0.218 for the constant-mode oracle. Because the rolling method is causal and uses only a trailing window whereas the oracle uses the entire record, matching the oracle is a result in favour of the rolling method. The median onset lead is negative on every usable channel, indicating that the count crossing tends to precede the catalog begin time.

Table 3. Proton trigger, rolling ($W = 30$ d, $k = 10$) versus the oracle fixed-threshold baseline, on the OU/OUT channels, scored against the NOAA proton catalog. Subscripts: r = rolling; p = oracle percentile mode; c = oracle constant mode. “lead” is the median onset difference (KSEM onset minus NOAA begin; negative means KSEM leads). PD1A-OU is an automatically flagged noise channel and is excluded from the mean.

Channel	POD _{r}	FAR _{r}	lead (h)	POD _{p}	FAR _{p}	POD _{c}	FAR _{c}
PD3A-OU	0.735	0.103	−1.00	0.735	0.143	0.676	0.551
PD3A-OUT	0.676	0.120	−0.92	0.706	0.115	0.706	0.452
PD1A-OUT	0.588	0.136	−0.67	0.588	0.174	0.588	0.174
PD1B-OU	0.559	0.100	−0.50	0.529	0.150	0.559	0.100
PD2B-OUT	0.559	0.100	−0.67	0.559	0.100	0.559	0.143
PD1B-OUT	0.559	0.100	−0.67	0.559	0.100	0.559	0.143
PD2A-OU	0.559	0.143	−0.50	0.559	0.182	0.559	0.308
PD2A-OUT	0.559	0.143	−0.67	0.559	0.143	0.559	0.143
PD3B-OU	0.559	0.100	−0.50	0.559	0.143	0.559	0.143
PD2B-OU	0.559	0.143	−0.58	0.529	0.150	0.559	0.143
PD3B-OUT	0.559	0.100	−0.67	0.559	0.100	0.559	0.100
<i>PD1A-OU (noise)</i>	0.412	0.856	+0.08	0.471	0.904	0.588	0.825
mean (excl. PD1A-OU)	0.591	0.112		0.586	0.136	0.586	0.218

Table 4. Main ablation: one design axis varied from the proposed configuration (**quietoff** / **MAD** / $W = 30$ d / $k = 10$). OU/OUT means exclude PD1A-OU.

Configuration	POD	FAR	PD3A-OU POD	PD3A-OU FAR
Proposed: quietoff / MAD / W30 / k10	0.591	0.112	0.735	0.103
σ : MAD $\rightarrow$ std	0.259	0.137	0.265	0.100
quiet: full $\rightarrow$ quiet7	0.591	0.124	0.735	0.107
rolling $\rightarrow$ fixed oracle (percentile)	0.586	0.136	0.735	0.143
rolling $\rightarrow$ fixed oracle (constant)	0.586	0.218	0.676	0.551

The method also identifies a hardware-noisy channel automatically rather than by hand. PD1A-OU has the highest FAR of any OU channel (0.856), a quiet-background zero-count fraction of 6.6%, and a standard-deviation-to-MAD ratio of 1,524—the signature of an intermittent, spiky series rather than a smooth particle flux—and is therefore excluded from the channel mean.

The main ablation (Table 4) varies one design axis at a time from the proposed configuration. The largest effect by far is the dispersion estimator: replacing MAD with the sample standard deviation collapses POD from 0.591 to 0.259, confirming that a robust scale is essential for the overdispersed count channels. The quiet-day selection and the rolling-versus-fixed change move FAR by a few points without changing POD. The constant- C baseline is the worst on FAR (0.218), because a single per-logic constant cannot adapt to per-channel background differences.

4.3 Electron trigger

No configuration achieves a usable electron trigger. Table 5 reports rolling and oracle performance on the FTU channels against the SWPC electron catalog; every configuration has $\text{FAR} \geq 0.63$. The apparently high POD of the constant-mode baseline

Table 5. Electron trigger, rolling versus oracle, on the FTU channels, scored against the SWPC electron catalog. No configuration achieves a usable trigger: all have FAR ≥ 0.63 .

Channel	POD _r	FAR _r	POD _p	FAR _p	POD _c	FAR _c
PD2B-FTU	0.550	0.637	0.183	0.853	0.708	0.658
PD2A-FTU	0.525	0.685	0.183	0.851	0.633	0.696
PD3B-FTU	0.500	0.700	0.192	0.832	0.775	0.627
PD1A-FTU	0.517	0.692	0.175	0.841	0.642	0.663
PD1B-FTU	0.475	0.676	0.150	0.839	0.500	0.680
PD3A-FTU	0.483	0.714	0.183	0.831	0.883	0.630
mean	0.508	0.684	0.178	0.841	0.690	0.659

Table 6. Trailing-window length and cold start (proton, `quietoff`/MAD, $k = 10$). Performance is flat from 5 to 30 days; the 30-day default gives the lowest FAR.

Window (d)	OU/OUT POD (mean)	OU/OUT FAR (mean)	PD3A-OU POD	PD3A-OU FAR
5	0.588	0.135	0.735	0.152
7	0.594	0.123	0.765	0.094
14	0.591	0.119	0.765	0.125
30 (default)	0.591	0.112	0.735	0.103

is an artefact of a fixed constant that happens to catch part of the population at an unusable false alarm ratio. The physical reason for the failure—gradual radiation-belt enhancements on a variable background—is taken up in Section 5.2.

4.4 Trailing-window length and cold start

Proton performance is flat across trailing windows from 5 to 30 days (Table 6), with the 30-day default giving the lowest FAR. The trigger therefore tolerates windows as short as 5 days, so the cold-start interval the spacecraft must accumulate after deployment is days rather than months.

4.5 Threshold multiplier sensitivity

POD is flat across $k = 3$ –10 and FAR decreases mildly over that range, with $k = 10$ giving the minimum FAR (Table 7); we therefore describe $k = 5$ –10 as a stable plateau with $k = 10$ as its lowest-FAR point rather than a sharp optimum. Notably, $k = 10$ improves on the standard 3σ choice ($k = 3$) in FAR (0.112 vs. 0.127) at equal POD. Beyond the plateau, increasing k raises the threshold and starves detection: POD declines by $k = 30$ and collapses at the σ -dominated extreme ($k = 999$, shown only to bound the trend). The $k\sigma$ term is what suppresses false alarms relative to $k = 0$ (median only), which keeps POD but raises FAR.

4.6 Detection floor sensitivity and on-board cost

The post-filter floor `peak_floor` trades detection against false alarms monotonically (Table 8): lowering it raises both POD and FAR, raising it lowers both. `peak_floor = 2` is the knee of the trade-off, where FAR has dropped to about 0.11 while POD is still 0.59; below it FAR rises steeply and above it POD falls without further FAR benefit. The method is robust within the 2–3 range, which matters for instrument transfer: a detector with

Table 7. Threshold multiplier k sensitivity (`quietoff`/MAD, $W = 30$ d).

Configuration	POD	FAR	PD3A-OU POD	PD3A-OU FAR
$k = 0$ (median only)	0.591	0.150	0.735	0.250
$k = 3$ (standard 3σ)	0.591	0.127	0.735	0.143
$k = 5$	0.591	0.117	0.735	0.103
$k = 10$ (proposed)	0.591	0.112	0.735	0.103
$k = 20$	0.588	0.126	0.735	0.150
$k = 30$	0.564	0.128	0.559	0.154
$k = 999$ (σ -dominated)	0.053	0.460	0.059	0.333

Table 8. `peak_floor` sensitivity (`quietoff`/MAD, $W = 30$ d, $k = 10$).

<code>peak_floor</code>	OU/OUT POD	OU/OUT FAR	PD3A-OU POD	PD3A-OU FAR
0.5	0.781	0.396	0.853	0.461
1	0.725	0.223	0.853	0.386
2 (proposed)	0.591	0.112	0.735	0.103
3	0.521	0.120	0.647	0.083

a different absolute count scale would re-tune `peak_floor`, and the flat 2–3 FAR behaviour indicates the method tolerates that adjustment.

The on-board cost of the realization is measured directly on a Jetson Orin edge platform, replaying the KSEM count stream at flight cadence over the monitored OU/OUT channel set (Table 9). The single quantity that responds to window length is the background update `wp_eval`, which falls to about half (0.52) when the window is shortened from 30 to 7 days; the decision engine `fsm_infer`, the transport stages, and the power draw are unchanged between settings. The full chain completes in under 10 ms at the median, against a 15-minute sample cadence. The operational reading of these numbers is developed in Section 5.1.

4.7 LEO transfer validation (POES MetOp-03 and NOAA-19)

The preceding results characterise the trigger on the geostationary KSEM record. We now apply the identical detection engine (Section 3.8) to the two LEO detectors of Section 2.5, evaluated against the same NOAA proton and SWPC electron catalogs over 2019–2025 (42 and 159 events). The purpose is twofold: to test that the GEO-developed trigger transfers to the more structured LEO background, and to ask whether the static count statistics of a channel and its dynamic trigger performance identify the same channel—the LEO counterpart of the static-vs-dynamic theme of Section 3.1.

4.7.0.1 Proton onset detection transfers, and is reproducible across two spacecraft.

Table 10 reports per-channel POD/FAR against the NOAA proton catalog for both detectors at $W = 30$ d, ranked by the separation score $J' = \text{POD} - \text{FAR}$. On both spacecraft the single best trigger channel by J' is the omnidirectional proton channel `omni.p6` (MetOp-03 POD 0.738 / FAR 0.355, $J' = 0.383$; NOAA-19 POD 0.619 / FAR 0.432, $J' = 0.187$), and the same channel ordering recurs on both detectors despite their different ages, which is the reproducibility a detector-physics-driven result should show. The lower J on NOAA-19 is consistent with its older detector: the 2009 instrument carries more accumulated proton-fluence degradation than the 2018 MetOp-03 (Section 2.5; Sandanger et al. (2015)), which raises its background count level and so its false-alarm ratio, while

Table 9. On-board resource and latency, $W = 30$ d versus $W = 7$ d, on a Jetson-class edge platform, replaying the KSEM count stream at flight cadence over the monitored OU/OUT channels (PD3A-OU, PD3A-OUT, PD1A-OUT, PD2A-OU). Values are medians (or means, as labelled) over the active post-warm-up segment across 15 runs. **wp_eval**: rolling median+MAD update; **fsm_infer**: per-sample channel-set reduction and persistence check; **dds**: WP-to-AP transport; **action**: AP-to-SC transport. Power is VDD.IN via tegrastats.

Quantity	$W = 30$ d	$W = 7$ d	W7 / W30
wp_eval (ms, median)	7.31	3.72	0.52
wp_eval (ms, p95)	9.11	4.38	—
dds transport (ms, median)	1.32	1.32	1.00
fsm_infer (ms, median)	0.036	0.036	1.00
node latency (ms, median)	0.40	0.41	—
action transport (ms, median)	0.92	0.96	—
end-to-end chain (ms, median sum)	9.7	6.2	—
power VDD.IN (mW, mean)	4423	4371	0.99
GPU (% , mean)	14.9	15.5	—
CPU (% , mean)	4.6	4.2	—
RAM (MB, mean)	3167	3082	—

the relative-rise trigger keeps its detection rate broadly intact—NOAA-19 is therefore the harder of the two LEO cases rather than a qualitatively different one. The directional high-energy proton channels **pro_tel10/tel190** P4 reach the highest raw POD (0.69–0.86) but at a much higher FAR (0.64–0.68), while **pro_tel10.p5** sits at a moderate POD with the lowest SAA-associated false-alarm count of any candidate (**n_fa_saa** = 1–2). These are not a single “best” channel but a tier of use-cases: **omni.p6** as the balanced performance channel, **pro_tel10.p5** as the SAA-immune specialist, and the P4 channels as high-sensitivity/high-noise.

4.7.0.2 The channel favoured by static statistics is not the channel that triggers best. This is the central LEO finding. By the static event-versus-background count statistics (the condition profile of Section 4.1’s LEO counterpart), **pro_tel10.p5** is the cleanest proton channel on both spacecraft: its 99th-percentile count rises from about 0.07 in quiet intervals to roughly 20 during proton events (20.1 on MetOp-03, 20.4 on NOAA-19) while reaching only 8.6 (MetOp-03) to 13.9 (NOAA-19) in the SAA, so events exceed the SAA in this channel. By the same static measure **omni.p6** looks SAA-contaminated on both: its SAA 99th percentile (about 2175 on MetOp-03, 2160 on NOAA-19) exceeds its event 99th percentile (about 1280 and 1310 respectively), a 99th-percentile ratio of roughly 1.6–1.7 in favour of the SAA. A reading based on static statistics alone would therefore prefer **pro_tel10.p5** and distrust **omni.p6**. The dynamic trigger reverses this: **omni.p6** is the top channel by J' on both spacecraft, and its SAA-associated false fraction is in fact low (**n_fa_saa_frac** about 0.07–0.09). The reason, taken up in Section 5.4, is that the rolling-median background absorbs the periodic SAA enhancement into the threshold itself, so the SAA rarely produces a detected onset even on a channel whose static distribution is SAA-heavy. Static dispersion and dynamic trigger performance measure different things.

4.7.0.3 The proton channels are proton-selective. Table 12 contrasts the NOAA (proton) and SWPC (electron) matches for the candidate channels. **omni.p6** detects proton events well (POD 0.738 on MetOp-03) but essentially ignores the electron catalog (POD 0.025), and **pro_tel10.p5** behaves the same way (NOAA 0.571 vs. SWPC 0.031); both hold on NOAA-19. The proton trigger channels respond to proton events and not

Table 10. LEO proton trigger: per-channel POD/FAR on two POES detectors (NOAA catalog, $W = 30$ d, $k = 10$), ranked by $J' = \text{POD} - \text{FAR}$. **n_fa_saa** = false alarms whose onset is SAA-associated ($|B| < 25,000$ nT); **n_fa_saa_frac** = **n_fa_saa** / **n_fa**. Top channels by J' shown.

Channel	n_det	n_hit	n_fa	n_fa_saa	saa_frac	POD	FAR	J'
<i>MetOp-03</i>								
omni.p6	203	31	72	5	0.069	0.738	0.355	0.383
pro.tel0.p4	373	36	240	8	0.033	0.857	0.643	0.214
pro.tel90.p4	437	35	286	10	0.035	0.833	0.654	0.179
pro.tel90.p5	146	28	75	1	0.013	0.667	0.514	0.153
pro.tel0.p5	100	24	47	1	0.021	0.571	0.470	0.101
omni.p7	92	14	26	4	0.154	0.333	0.283	0.050
<i>NOAA-19</i>								
omni.p6	222	26	96	9	0.094	0.619	0.432	0.187
pro.tel90.p4	399	30	272	10	0.037	0.714	0.682	0.032
pro.tel0.p4	364	29	247	7	0.028	0.690	0.679	0.011
pro.tel90.p5	146	23	80	2	0.025	0.548	0.548	0.000
pro.tel0.p5	122	21	66	2	0.030	0.500	0.541	-0.041
omni.p7	95	13	38	6	0.158	0.310	0.400	-0.090

Table 11. Same engine, three detectors: proton-trigger POD/FAR across orbits. The KSEM (GEO) row is the best single proton channel (PD3A-OU) at $W = 30$ d (Table 6); the POES (LEO) rows are the best channel by J' (**omni.p6**; Table 10), also at $W = 30$ d. GEO requires channel selection only; LEO additionally applies SAA tagging (Section 3.8). POD/FAR are not strictly comparable across detectors because the catalogs are matched over slightly different spans and the channels differ physically; the table shows that the same engine produces a working proton trigger in both orbit regimes.

Detector	Orbit	Best proton channel	POD	FAR	SAA handling
KSEM / GK2A	GEO	PD3A-OU	0.735	0.103	none (no SAA)
POES MetOp-03	LEO	omni.p6	0.738	0.355	post-hoc tag
POES NOAA-19	LEO	omni.p6	0.619	0.432	post-hoc tag

to radiation-belt electron enhancements, even though the latter are far more numerous in LEO, confirming species selectivity.

4.7.0.4 Window length is immaterial, reproducing the KSEM cold-start result. Shortening the trailing window from 30 to 7 days leaves POD/FAR essentially unchanged on both spacecraft (e.g. MetOp-03 **omni.p6** POD $0.738 \rightarrow 0.690$, NOAA-19 **omni.p6** POD $0.619 \rightarrow 0.619$; if anything FAR improves slightly), mirroring the flat window dependence found for KSEM in Section 4.4 and reinforcing that the cold-start interval is days, now on a second orbit regime.

4.7.0.5 An aged-detector artefact is isolated, not mistaken for signal. On NOAA-19, the **omni.p6** channel shows a single extreme excursion (max count about 1.0×10^6) at magnetic latitude 56.5 deg and $|B| = 37,219$ nT on 2024-04-22—that is, at mid-latitude and outside the SAA contour, so it is not an SAA transit. This is consistent with the aged-proton-detector degradation expected of the 2009 spacecraft (Section 2.5; Sandanger et al. (2015)), and it appears on NOAA-19 but not on the younger MetOp-03. It does not propagate into the matched detections, but it is the kind of single-channel artefact the diagnostics flag rather than score as an event.

Table 12. Proton selectivity: NOAA (proton) versus SWPC (electron) match ($W = 30$ d). Each candidate channel detects proton events but not electron events, confirming species selectivity.

Detector	Channel	NOAA POD	NOAA FAR	SWPC POD	SWPC FAR
MetOp-03	omni_p6	0.738	0.355	0.025	0.936
MetOp-03	pro_tel0_p5	0.571	0.470	0.031	0.940
NOAA-19	omni_p6	0.619	0.432	0.025	0.928
NOAA-19	pro_tel0_p5	0.500	0.541	0.038	0.934

5 Discussion

5.1 The trigger as an on-board gating layer

The results of Section 4.2 show that a causal rolling-background trigger matches an oracle fixed threshold for proton events. The operational value of this depends on the trigger being cheap enough to run continuously on board, since its role in the PADO-SAT concept is to gate a heavier prediction model rather than to replace it. Three properties established here support that role: the computation is intrinsically light, its cost scales with quantities the designer controls, and direct measurement on an edge platform confirms both at flight cadence.

First, the trigger’s computation is intrinsically light. Per update it requires only a median and a median absolute deviation over a trailing buffer, both order- n operations on a per-channel buffer of at most a few thousand samples, with no floating-point-intensive model evaluation. The state carried between updates is a single bounded ring buffer per channel. This contrasts with the convolutional or recurrent inference that an on-board prediction model would perform, and is what allows the trigger to run at every cadence while the expensive model runs only when gated.

Second, the resource cost scales with the two quantities the designer controls: the trailing-window length and the number of monitored channels. The median/MAD re-computation grows with window length, and the per-channel buffers and updates grow linearly with the channel count. Section 4.4 showed that the window can be shortened from 30 to as little as 5 days with negligible loss of proton detection; together with the channel diagnostics of Section 4.2, which justify monitoring only the few suitable channels rather than all of them, this gives the designer direct control over the trigger’s resource envelope.

Third, on-board measurement confirms these properties quantitatively. We replayed the KSEM count stream at flight cadence on a Jetson Orin edge platform over the monitored OU/OUT channel set, for both $W = 30$ d and $W = 7$ d and across the five event windows of Section 4, and recorded per-update inference time, node latency, alert-to-command transport, power draw (VDD_IN via tegrastats), and CPU/GPU/RAM usage; the active-segment results are reported in Table 9.

The measurements bear out the structural argument and sharpen it in one respect. The decision engine itself—the per-sample channel-set reduction and persistence check that turns a background update into an alert (Section 3.7)—costs a median of 0.036 ms per update, two orders of magnitude below the background update that precedes it. The full on-board chain, from a count sample arriving to a mode command being issued (background update, WP-to-AP transport, decision, AP-to-SC transport), completes in under 10 ms at the median for the heavier $W = 30$ d setting and in about 6 ms for $W = 7$ d. Against the 15-minute sample cadence at which the trigger operates, this is roughly

five orders of magnitude of headroom, which is the concrete content of the “lightweight enough for continuous triggering” claim: the trigger’s duty cycle on the processor is negligible, leaving the platform free for the gated prediction model the trigger exists to invoke. Power draw is about 4.4 W on this platform and, like the decision and transport stages, is effectively unchanged between the two window settings.

The one quantity that responds to window length is the background update itself. Shortening the trailing window from 30 to 7 days roughly halves its per-update time ($0.52\times$; 7.31 to 3.72 ms median), rather than the near-linear reduction the buffer size alone would suggest—the median and MAD are order- $n \log n$ operations with a fixed per-update overhead, so wall-clock time falls more gently than the sample count. This distinction matters for how the window trade-off should be read. The memory footprint does scale linearly with the window, so a shorter window buys a proportional reduction in per-channel buffer size; but the recurring compute cost, and the power draw, do not fall proportionally, and the power draw barely moves at all. The case for the shorter window is therefore one of memory and compute economy on a constrained processor, not of energy: Section 4.4 establishes that detection performance is flat down to a 5–7 day window, so a designer who is memory- or compute-bound can trade down to a 7-day window and recover roughly half the background-update cost and a proportional share of the buffer memory, at no measured cost in proton detection and no meaningful change in power.

Taken together, the cold-start tolerance (Section 4.4) and the measured computational lightness give two distinct operational advantages: the trigger becomes usable within days of deployment, and it can run continuously within a small resource budget thereafter. Both are prerequisites for the event-driven autonomy that motivates the mission. We note that these measurements are taken on a ground edge platform with the count stream replayed at cadence; they characterize the trigger’s intrinsic cost rather than a flight-qualified figure, and the in-orbit demonstration of the broader mission is where that cost is to be confirmed on the target processor and radiation environment.

5.2 Electrons: why a different detection principle is needed

The electron result of Section 4.3 is a negative one, and understanding why points to the remedy. The threshold-crossing detector is matched to the abrupt onset of a proton event, where the count rises sharply from a low, stable background. Radiation-belt electron enhancements, as recorded in the SWPC catalog, instead build up gradually on top of a high and strongly modulated background, so the moment a fixed or rolling level is crossed carries little information about a well-defined onset, and the same level is crossed frequently by background variation, inflating false alarms.

The established tools for exactly this regime are change-point methods rather than level-crossing methods. The Poisson-CUSUM scheme (Huttunen-Heikinmaa et al., 2005) accumulates deviations from an expected background and signals when the accumulated evidence, not the instantaneous value, exceeds a control limit; its bootstrap hybrid extension additionally quantifies onset-time uncertainty (Palmroos et al., 2025). These methods are more sensitive to a slow, sustained rise than a threshold is, which is the failure mode we observe for electrons. They are, however, more computationally involved and are designed for calibrated flux, so adapting a count-based, on-board-feasible change-point detector is the natural next step rather than a direct drop-in. A slope- or CUSUM-based electron front end, combined with the proton trigger demonstrated here, is left as future work; the contribution of the present result is to delimit precisely where the simple trigger applies and where it does not, rather than to obscure the boundary.

5.3 Limitations

Three limitations bound the claims of this paper.

First, although the trigger has now been validated on two LEO detectors (Section 4.7) and not only on the geostationary development instrument, that validation is an offline replay rather than a closed-loop, in-orbit demonstration. The LEO transfer establishes that the identical engine detects proton onsets on the structured POES background and that the South Atlantic Anomaly handling of Section 3.8 behaves as intended, but the South Atlantic Anomaly tagging is evaluated post hoc rather than in a live control loop, and the timing was measured by replaying the count stream at cadence rather than on a flying processor. The remaining gap is therefore not the orbit regime, which is now covered by GEO development plus LEO validation, but the step from ground replay to in-flight execution; that step is the role of the in-orbit demonstration in the broader mission. This gap is neither hypothetical nor unique to the present work: the FIREBIRD-II CubeSat developed an on-board algorithm to identify rare microburst events autonomously, tested it on the ground, and found that it failed in flight on raw, unprocessed data, forcing a retreat to a scientist-in-the-loop scheme in which downlink intervals were chosen by hand from a coarse data product (Spence et al., 2022). That a ground-validated on-board selector can fail in flight is precisely why we frame the present LEO result as a replay-based characterization and defer the closed-loop claim. Conversely, the goal of conveying event onset information in real time has a working precedent: the REPT space-weather beacon downlinked a few selected raw integral counting-rate channels continuously, providing real-time situational awareness of radiation-belt activity without ground processing (Baker et al., 2021). The contribution here is not real-time operation per se, which has been demonstrated, but the specific combination of a flux-free, instrument-agnostic finite-state trigger validated across two orbit regimes. A secondary caveat within the LEO validation is that the older NOAA-19 proton detector exhibits the accumulated degradation expected of a 2009 instrument (Section 4.7); because the trigger keys on a relative rise above a rolling background rather than an absolute count level, this degradation affects the absolute count scale more than the trigger decision, but it is a reason to read the NOAA-19 numbers as the harder of the two LEO cases.

Second, the evaluation period (2019–2024 for KSEM, extended to 2025 for the LEO catalogs) covers the rising phase and maximum of solar cycle 25 but no deep minimum. A fixed oracle threshold benefits from a relatively stationary background over such a period, so the close match between the rolling method and the oracle baseline (Section 4.2) may understate the rolling method’s advantage: over a full cycle, with larger background drift, the adaptive threshold should pull further ahead of any fixed value. The comparison reported here is thus conservative with respect to the proposed method.

Third, the ground-truth catalogs carry their own uncertainties. The matching tolerance of plus or minus 24 hours absorbs the imprecision of catalog onset and end times, but it also means the reported onset lead of the KSEM counts (Section 4.2) should be read as a central tendency rather than an event-by-event guarantee. The electron catalog’s daily-maximum peak convention is coarser still, which compounds the physical mismatch discussed in Section 5.2 and is a further reason the electron result is reported conservatively.

5.4 Static count statistics are not dynamic trigger performance

The LEO results of Section 4.7 sharpen the static-versus-dynamic theme of Section 3.1 into a more operational distinction. The theme is that a channel’s static count separability does not predict its value as a trigger. The POES proton channels show this gap directly, between two properties that are both intrinsic to the count series: the static separation of a channel’s count distribution and its dynamic performance once that channel drives the rolling-background detector.

The clearest case is the omnidirectional channel `omni_p6`. Judged by static event-versus-background statistics it looks unsuitable: its count distribution during South At-

lantic Anomaly transits reaches higher values than during the proton events themselves (its SAA 99th percentile exceeds its event 99th percentile by a factor of roughly 1.6–1.7 on both spacecraft), exactly the signature one would reject as SAA-contaminated. Judged by the dynamic trigger it is the best proton channel on both spacecraft, with the highest J' ($= \text{POD} - \text{FAR}$) and an SAA-associated false-alarm fraction of under ten percent. The channel that the static statistics most distrust is the one that triggers best.

The resolution is that the rolling-median background, the very mechanism the trigger is built on, treats the SAA differently from a static percentile. The SAA enhancement is periodic: every transit raises the count, so the trailing median rises with it and the threshold tracks the recurring enhancement rather than being repeatedly crossed by it. A static percentile, computed once over the whole record, instead sees the SAA peaks as part of one pooled distribution and is dominated by them. The dynamic threshold thus absorbs the structured, repeating component of the background that the static statistic can only treat as noise. This is why `omni_p6`'s SAA-driven false fraction is low in practice despite its SAA-heavy static distribution, and why a channel-selection rule based on static separation alone would have discarded the best trigger.

The practical consequence for on-board channel selection is concrete: channels should be ranked by their performance under the actual detector, against a catalog, rather than by static distributional cleanliness. The two diagnostics are complementary, not interchangeable. The static profile remains useful for identifying a genuinely hardware-noisy channel—as with the automatically flagged KSEM noise channel of Section 4.2 and the aged-detector artefact on NOAA-19—but it is the dynamic, catalog-referenced evaluation that must decide which channels a mission actually monitors. The complementary pairing also suggests a multi-channel operating point: `omni_p6` as the balanced performance channel and a directional channel such as `pro_tel0_p5`, whose SAA-associated false count is the lowest of any candidate, as an SAA-immune cross-check, combined through the reverse-Polish voting logic already provided by the on-board architecture of Section 3.7.

6 Conclusion

We have evaluated whether solar energetic particle onsets can be detected from raw, uncalibrated particle counts well enough to serve as the low-power trigger that gates on-board inference in an event-driven CubeSat. Using five and a half years of KSEM/GK2A logic-channel counts and two independent event catalogs, we tested an instrument-agnostic finite-state detector built on a rolling robust (median + MAD) background, against a fixed-threshold oracle baseline tuned with full multi-year hindsight. We then transferred the identical detection engine, without retuning, to two low-Earth-orbit particle monitors (POES MetOp-03 and NOAA-19) and evaluated it against the same catalogs.

Four findings follow. For protons on the geostationary development instrument, the causal rolling trigger matches the oracle fixed threshold (mean POD 0.591 at FAR 0.112 on the OU/OUT channels), establishing that no pre-accumulated archive is needed for on-board use and that the cold-start interval is days rather than months. For electrons, the threshold-crossing principle fails under both fixed and rolling thresholds, not for algorithmic reasons but because radiation-belt enhancements rise gradually on a variable background, delimiting precisely where the simple trigger applies and where it does not. Across both species on KSEM, the channels whose static count distributions look cleanest are not the channels that best trigger, and a single noisy channel is identified automatically by the method, giving a concrete criterion for on-board channel selection. On LEO, the trigger detects proton onsets reproducibly on both spacecraft—`omni_p6` is the top channel by J' ($= \text{POD} - \text{FAR}$) on MetOp-03 (POD 0.738, FAR 0.355, $J' = 0.383$) and NOAA-19 (POD 0.619, FAR 0.432, $J' = 0.187$)—and the channel favoured by static count statistics (`pro_tel0_p5`, whose event counts exceed its SAA counts) is not the channel that triggers best (`omni_p6`, whose periodic SAA enhancement is absorbed into the

rolling background rather than generating false alarms), the same static-vs-dynamic split confirmed in the LEO environment.

The broader implication is methodological rather than algorithmic. The individual components—rolling robust background estimation, catalog-based evaluation, and per-channel diagnostics—are each standard; their value here is in being combined and characterized for the specific problem of a flux-free, cold-start-tolerant, channel-agnostic on-board trigger, a combination for which no prior end-to-end validation existed across two orbital regimes. That the dispersion estimator dominates the result, with a robust MAD scale being essential where the sample standard deviation collapses detection, is the most transferable empirical lesson.

The trigger was developed on a geostationary instrument (KSEM) and validated on two LEO detectors (POES MetOp-03 and NOAA-19), detecting proton onsets reproducibly across both orbit regimes with the identical engine and operating points. The evaluation spans the rising phase and maximum of solar cycle 25 but no deep minimum, which makes the comparison with the oracle baseline conservative with respect to the adaptive method. The natural next steps are a change-point (CUSUM-style) front end for the electron regime, and an in-orbit, closed-loop demonstration on the target processor and radiation environment. Within the PADO-SAT concept, the trigger characterized here is the rule-based layer that precedes and gates the on-board prediction model, and the present results—spanning a geostationary development phase and a LEO validation phase—support its feasibility as that layer.

Open Research Section

KSEM/GK2A particle-detector Level-0 count data are provided by the National Meteorological Satellite Center (NMSC) of the Korea Meteorological Administration. The NOAA GOES solar proton event list and the SWPC ALTEF3 electron alert list are available from NOAA SWPC. POES/MetOp SEM-2 (MEPED) count data are available from NOAA NCEI. The analysis code implementing the finite-state trigger and the evaluation pipeline is available at <https://github.com/saikey-gintama/Pixhawk.ReID/>.

Conflict of Interest declaration

The authors declare there are no conflicts of interest for this manuscript.

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